✅ EJERCICIOS BÁSICOS (CACTI + XYMON)

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1. Verificar instalación inicial de Cacti (básico)

Enunciado

Tras instalar Cacti, comprueba que Apache, MariaDB y SNMP están funcionando correctamente. Indica los comandos usados.

Solución

systemctl status apache2

systemctl status mariadb

systemctl status snmpd

Explicación

Cacti depende directamente de estos tres servicios:

- Apache sirve la interfaz web.

  sudo systemctl start apache2

sudo systemctl enable apache2

sudo systemctl status apache2

- MariaDB guarda la información.

  sudo systemctl start mariadb

  sudo systemctl enable mariadb

  sudo systemctl status mariadb

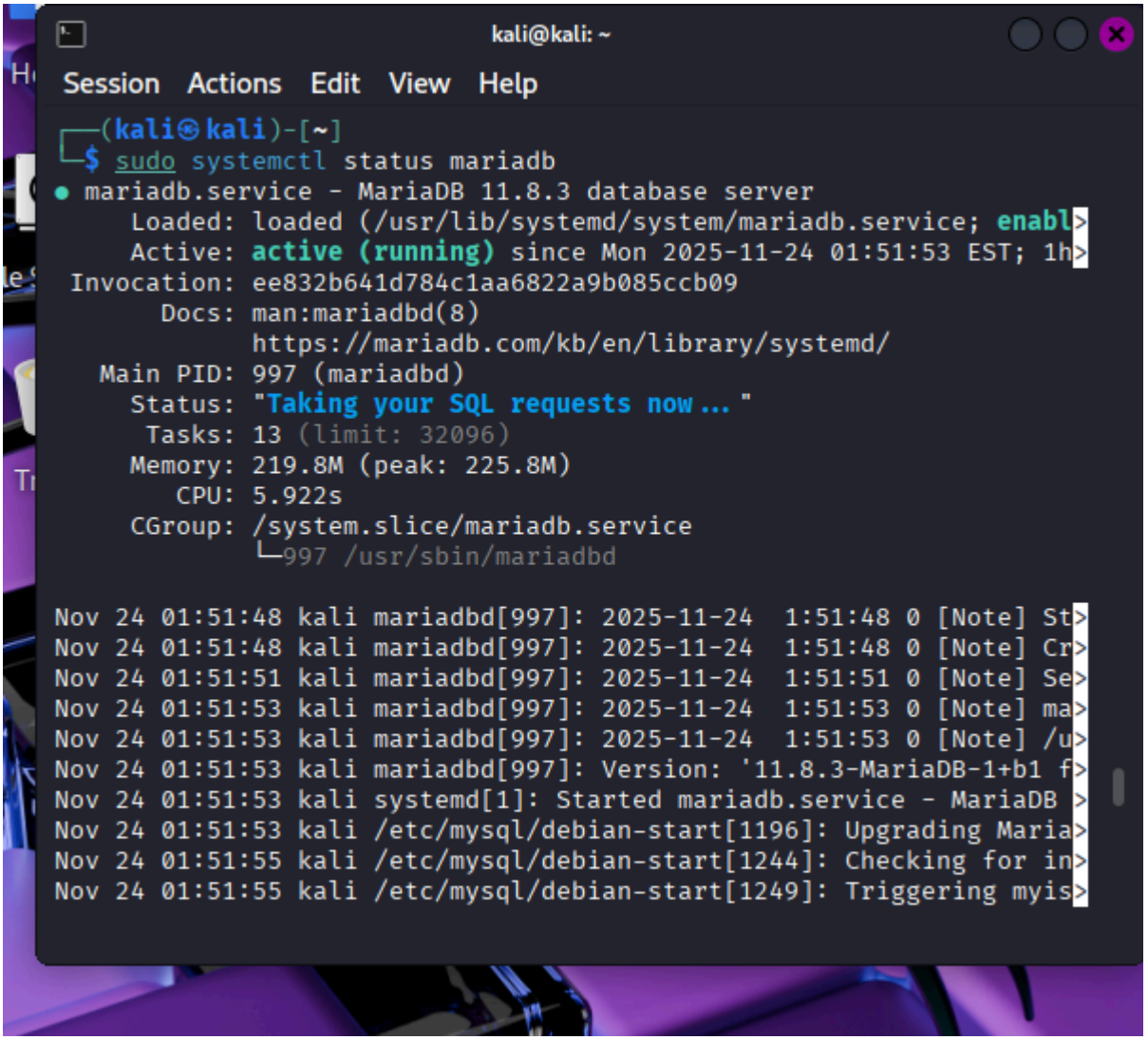
- SNMP es la fuente de métricas.
  Tu documento menciona explícitamente instalarlos y reiniciarlos.

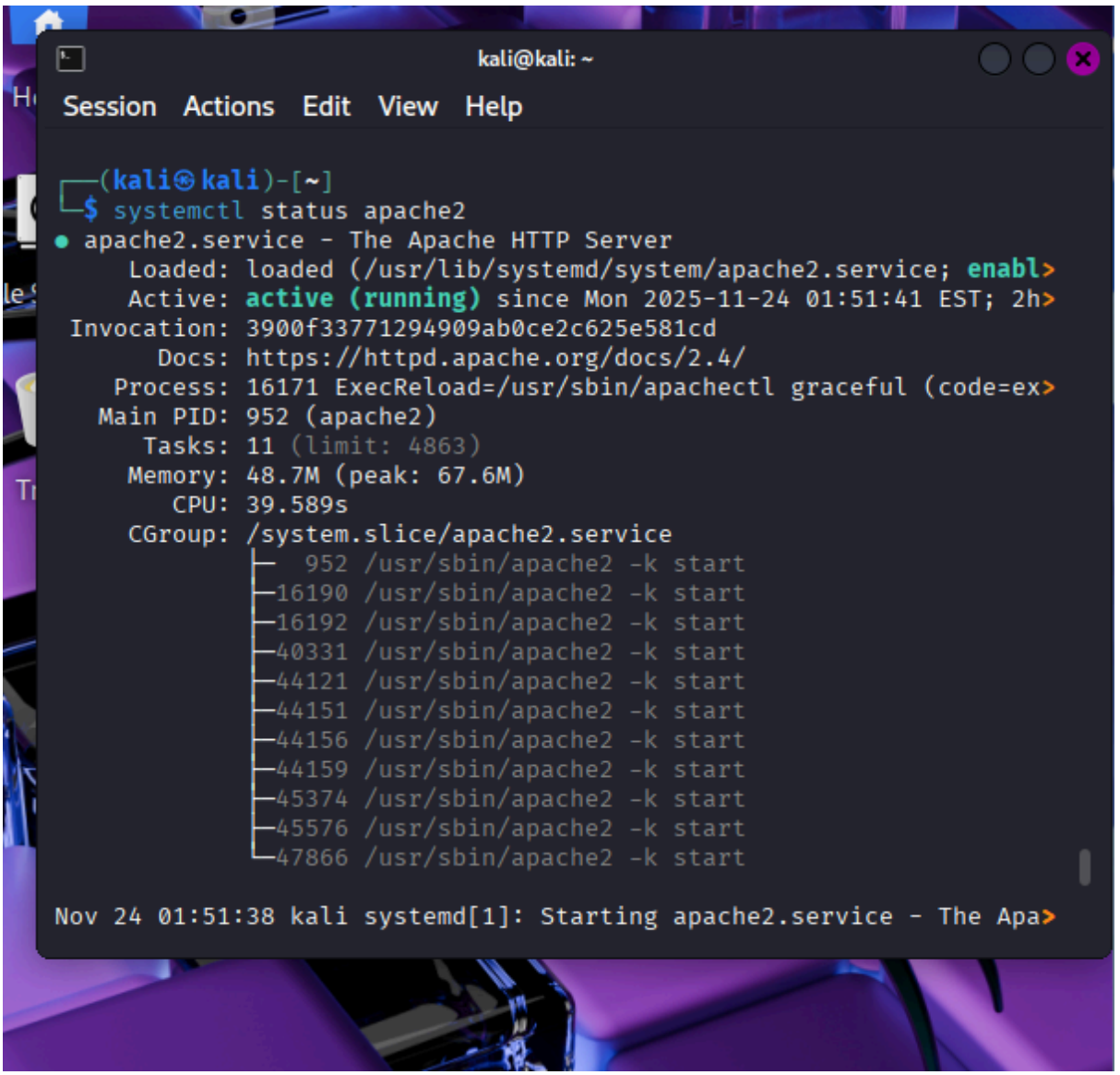
  sudo apt update

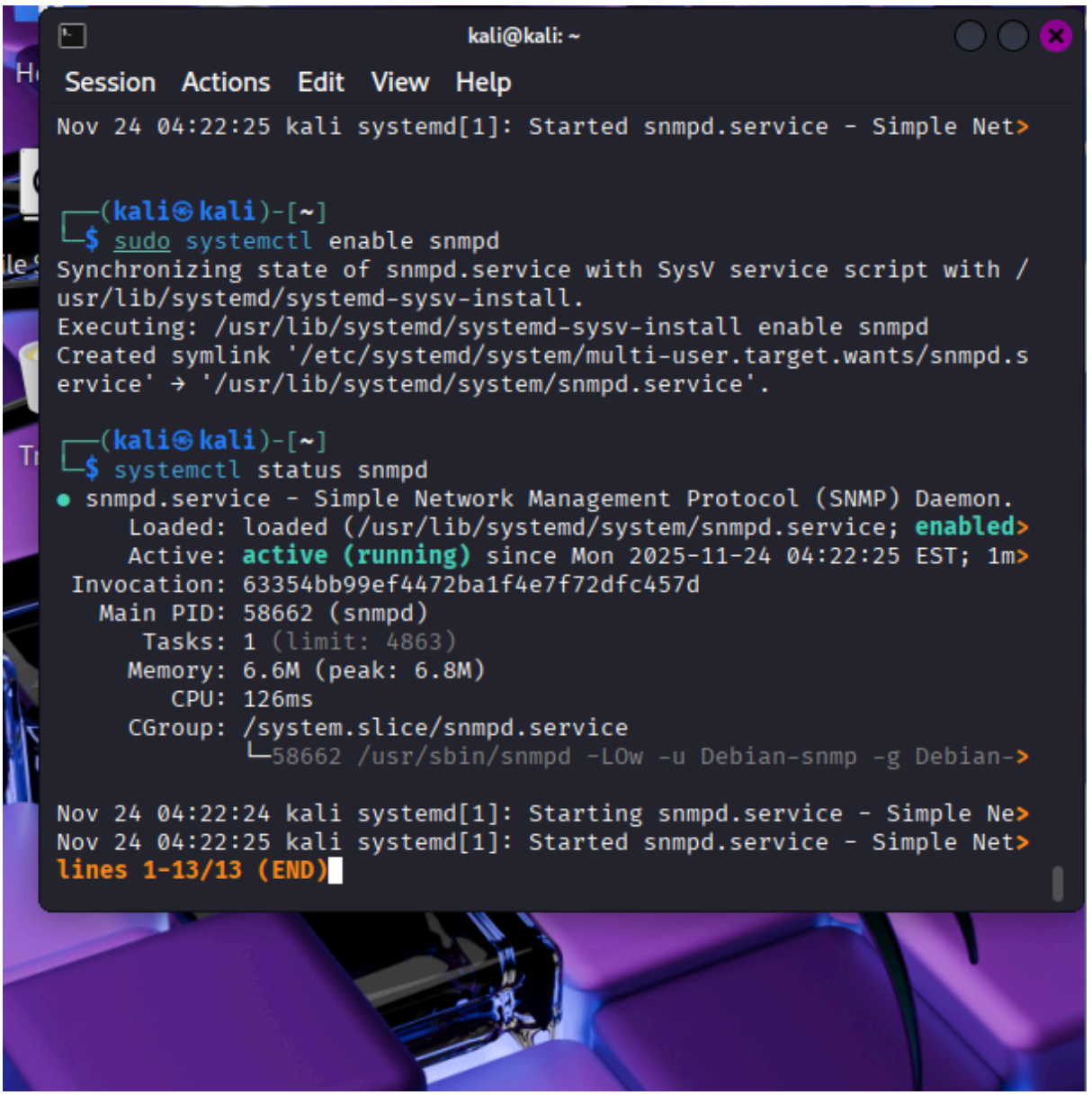
  sudo apt install snmpd

  sudo systemctl enable snmpd

  sudo systemctl start snmpd


kali@kali: ~
Session Actions Edit View Help
(kali㉿kali)-[~]
$ sudo systemctl status mariadb
● mariadb.service - MariaDB 11.8.3 database server
Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabl>
Active: active (running) since Mon 2025-11-24 01:51:53 EST; 1h>
Invocation: ee832b641d784c1aa6822a9b085ccb09
Docs: man:mariadbd(8)
https://mariadb.com/kb/en/library/systemd/
Main PID: 997 (mariadbd)
Status: "Taking your SQL requests now..."
Tasks: 13 (limit: 32096)
Memory: 219.8M (peak: 225.8M)
CPU: 5.922s
CGroup: /system.slice/mariadb.service
└997 /usr/sbin/mariadbd
Nov 24 01:51:48 kali mariadbd[997]: 2025-11-24 1:51:48 0 [Note] St>
Nov 24 01:51:48 kali mariadbd[997]: 2025-11-24 1:51:48 0 [Note] Cr>
Nov 24 01:51:51 kali mariadbd[997]: 2025-11-24 1:51:51 0 [Note] Se>
Nov 24 01:51:53 kali mariadbd[997]: 2025-11-24 1:51:53 0 [Note] ma>
Nov 24 01:51:53 kali mariadbd[997]: 2025-11-24 1:51:53 0 [Note] /u>
Nov 24 01:51:53 kali mariadbd[997]: Version: '11.8.3-MariaDB-1+b1 f>
Nov 24 01:51:53 kali systemd[1]: Started mariadb.service - MariaDB >
Nov 24 01:51:53 kali /etc/mysql/debian-start[1196]: Upgrading Maria>
Nov 24 01:51:55 kali /etc/mysql/debian-start[1244]: Checking for in>
Nov 24 01:51:55 kali /etc/mysql/debian-start[1249]: Triggering myis>


kali@kali: ~
Session Actions Edit View Help
(kali㉿kali)-[~]
$ systemctl status apache2
● apache2.service - The Apache HTTP Server
Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabl>
Active: active (running) since Mon 2025-11-24 01:51:41 EST; 2h>
Invocation: 3900f33771294909ab0ce2c625e581cd
Docs: https://httpd.apache.org/docs/2.4/
Process: 16171 ExecReload=/usr/sbin/apachectl graceful (code=ex>
Main PID: 952 (apache2)
Tasks: 11 (limit: 4863)
Memory: 48.7M (peak: 67.6M)
CPU: 39.589s
CGroup: /system.slice/apache2.service
├─ 952 /usr/sbin/apache2 -k start
├─16190 /usr/sbin/apache2 -k start
├─16192 /usr/sbin/apache2 -k start
├─40331 /usr/sbin/apache2 -k start
├─44121 /usr/sbin/apache2 -k start
├─44151 /usr/sbin/apache2 -k start
├─44156 /usr/sbin/apache2 -k start
├─44159 /usr/sbin/apache2 -k start
├─45374 /usr/sbin/apache2 -k start
├─45576 /usr/sbin/apache2 -k start
└─47866 /usr/sbin/apache2 -k start
Nov 24 01:51:38 kali systemd[1]: Starting apache2.service - The Apa>



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2. Acceder a la consola web de Xymon (básico)

Enunciado

Comprueba que Xymon está accesible desde el navegador utilizando la IP de la máquina virtual.

Solución

1. Obtener IP:

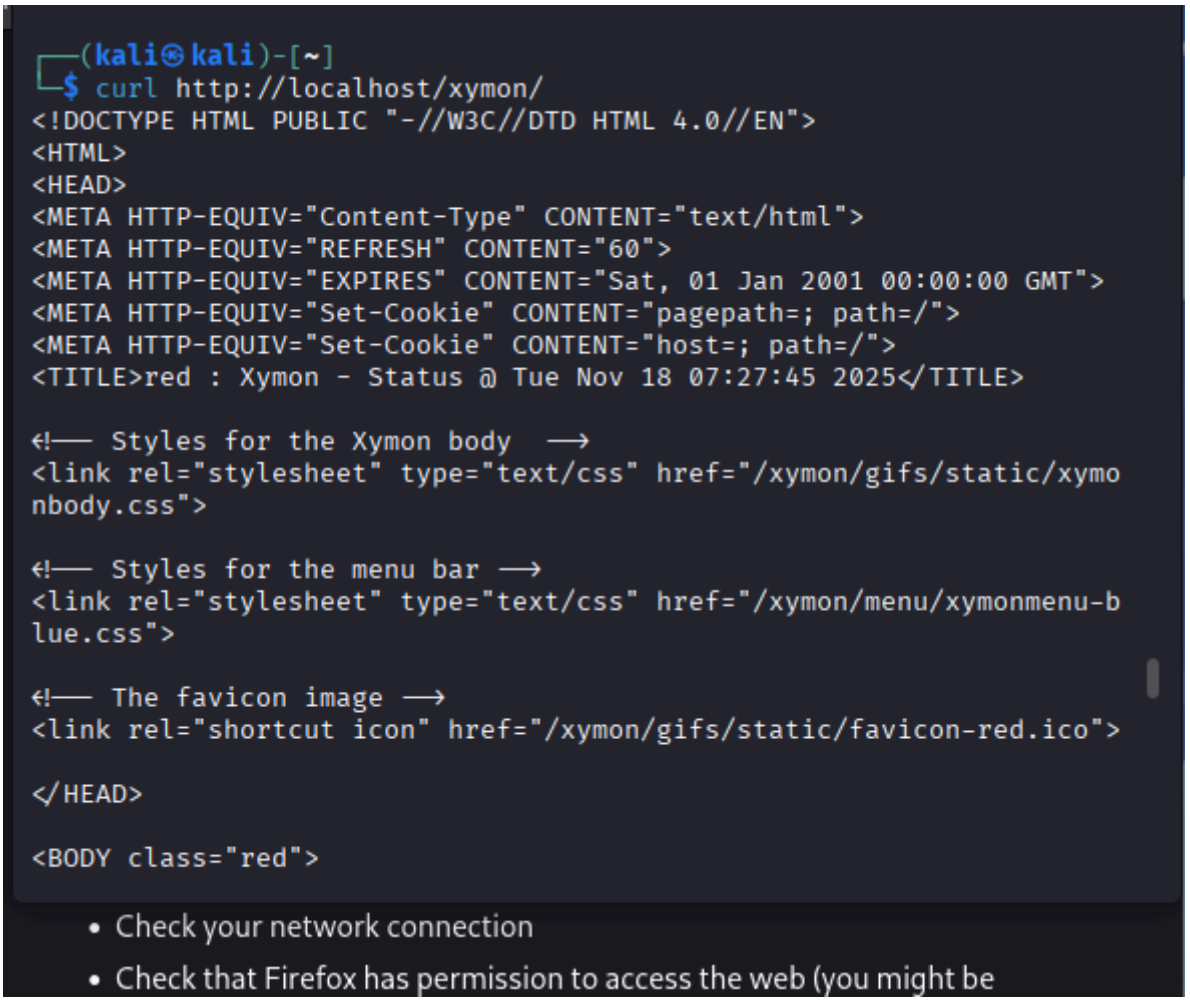
ip a

2. Probar acceso desde navegador:
   http://IP_DE_LA_VM/xymon/
3. Verificar desde consola:

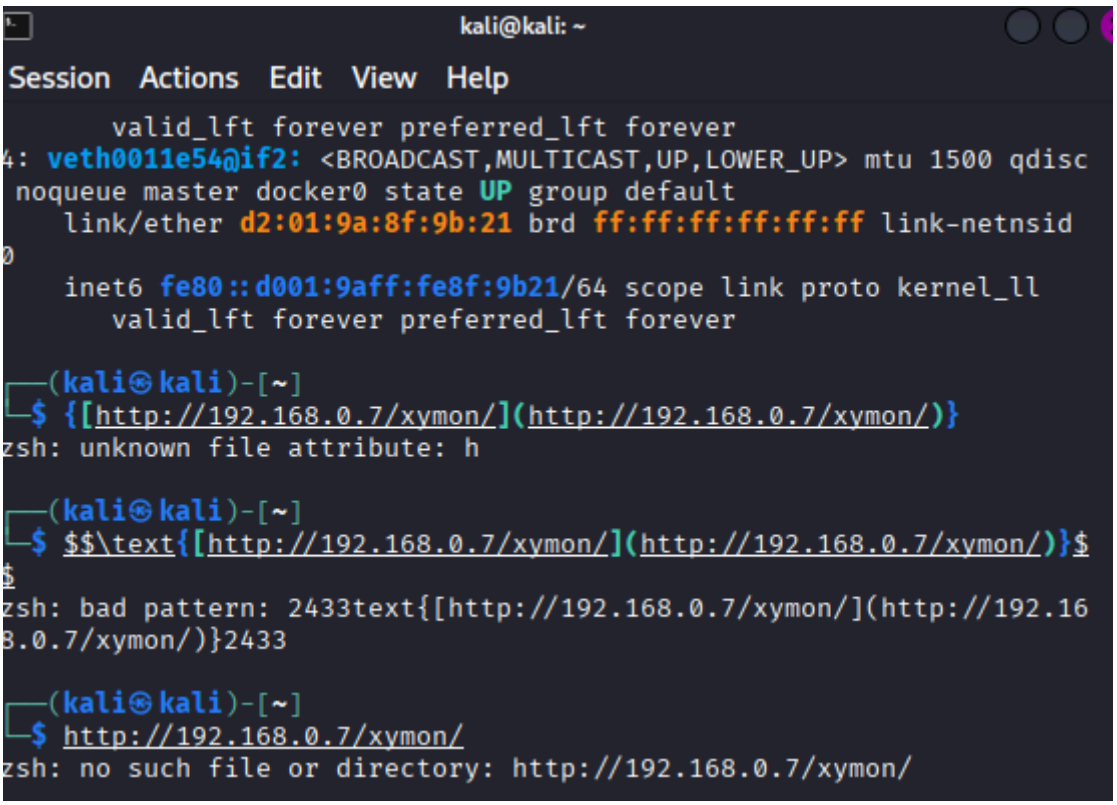
curl http://localhost/xymon/

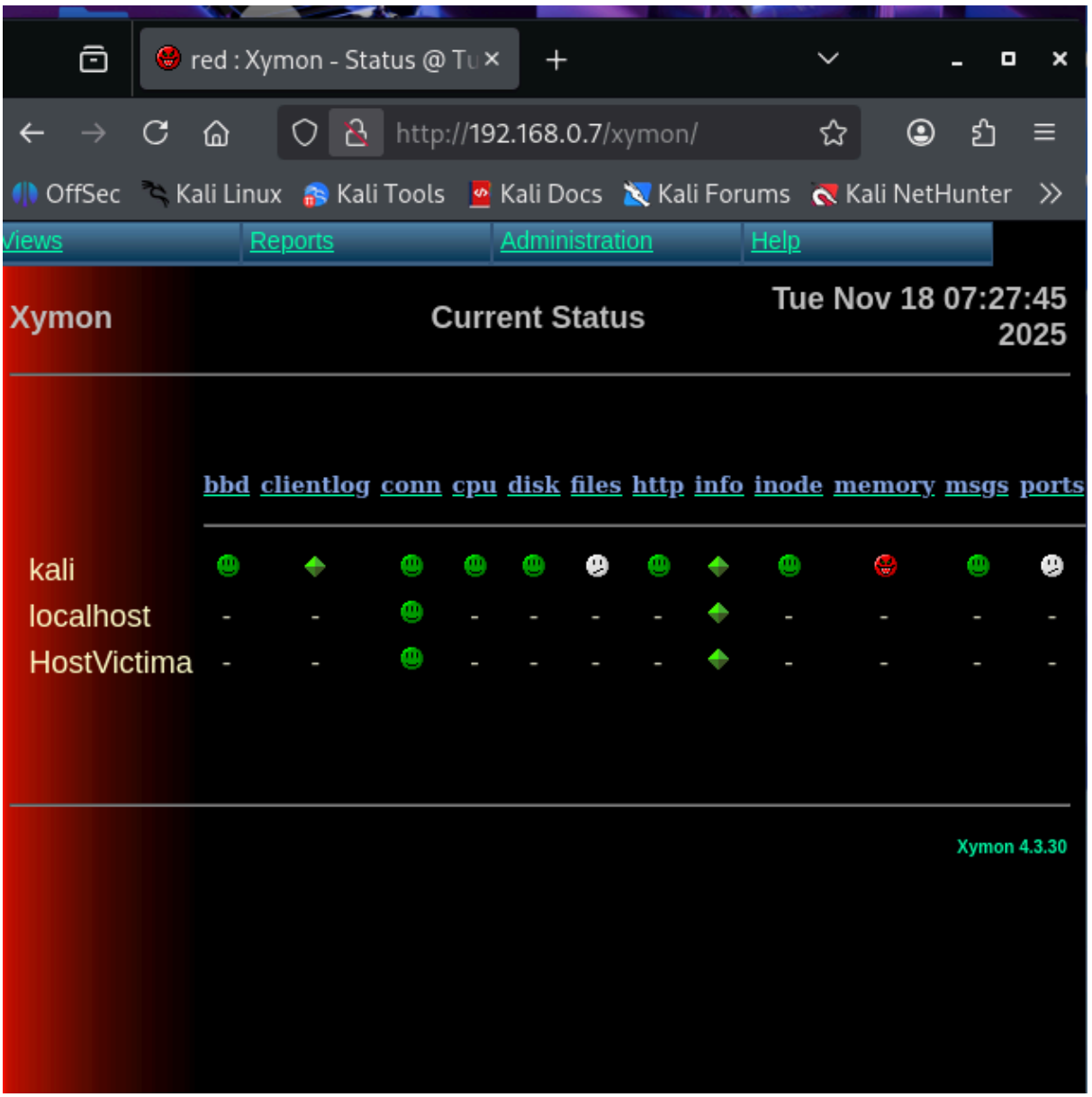
Explicación

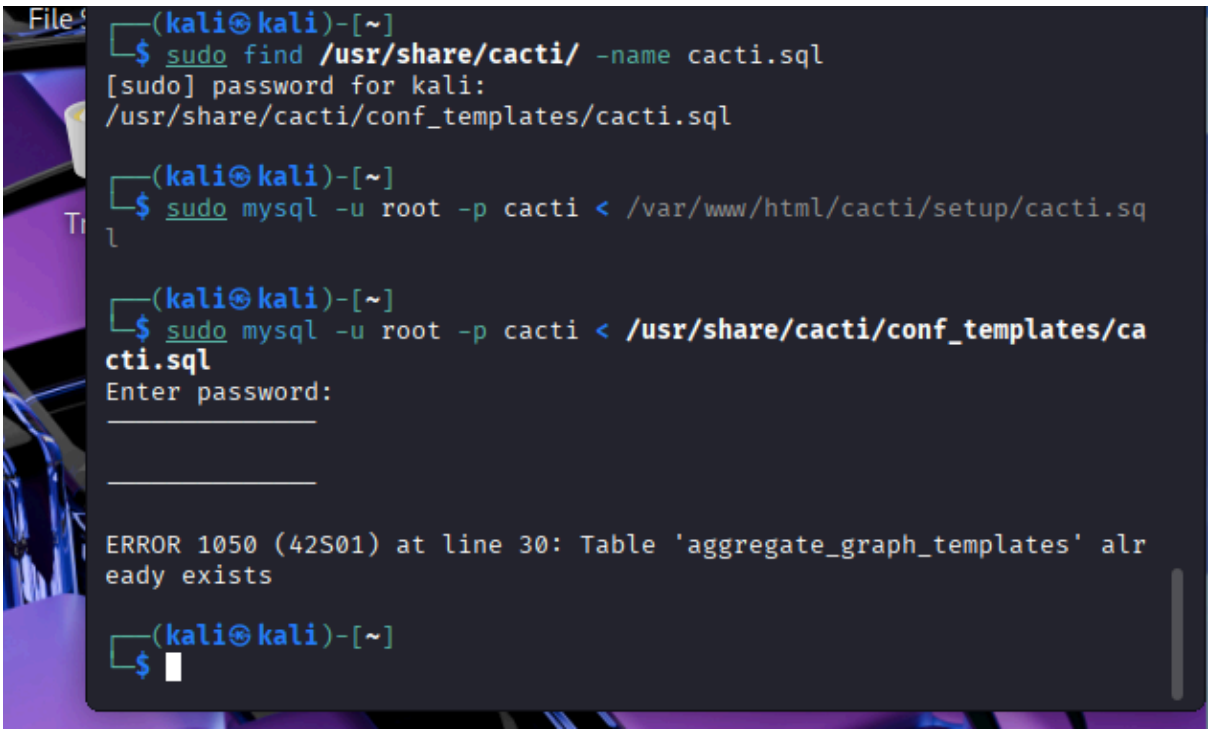
Tu documento lo indica: Xymon expone su dashboard en Apache y se accede vía /xymon.




kali@kali: ~
Session Actions Edit View Help
ver"><B><A HREF="http://xymon.sourceforge.net/" style="text-decorati
on: none">Xymon 4.3.30</A></B></FONT></TD>
</TR>
</TABLE>
<!-- Don't rely just on HTML to set the cookies -->
<!-- http://lists.xymon.com/pipermail/xymon/2018-June/045538.html --
>
<script type="text/javascript">
var metas = document.getElementsByTagName('meta');
for (var i=0; i<metas.length; i++) {
var type = metas[i].getAttribute("HTTP-EQUIV");
if (type == "Set-Cookie") {
var prop = metas[i].getAttribute("CONTENT");
document.cookie = prop;
}
}
</script>
</BODY>
</HTML>
(kali㉿kali)-[~]
$
• Check your network connection


kali@kali: ~
Session Actions Edit View Help
valid_lft forever preferred_lft forever
4: veth0011e54@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc
noqueue master docker0 state UP group default
link/ether d2:01:9a:8f:9b:21 brd ff:ff:ff:ff:ff:ff link-netnsid
0
inet6 fe80::d001:9aff:fe8f:9b21/64 scope link proto kernel_ll
valid_lft forever preferred_lft forever
┌──(kali㉿kali)-[~]
└─$ {http://192.168.0.7/xymon/}
zsh: unknown file attribute: h
┌──(kali㉿kali)-[~]
└─$ $$\text{http://192.168.0.7/xymon/}$
$
zsh: bad pattern: 2433text{[http://192.168.0.7/xymon/](http://192.16
8.0.7/xymon/)}2433
┌──(kali㉿kali)-[~]
└─$ http://192.168.0.7/xymon/
zsh: no such file or directory: http://192.168.0.7/xymon/


red : Xymon - Status @ Tu
http://192.168.0.7/xymon/
OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Views
Reports
Administration
Help
Xymon
Current Status
Tue Nov 18 07:27:45 2025
bbd clientlog conn cpu disk files http info inode memory msgs ports
kali
localhost
HostVictima
Xymon 4.3.30



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3. Importar la base de datos inicial de Cacti (básico)

Enunciado

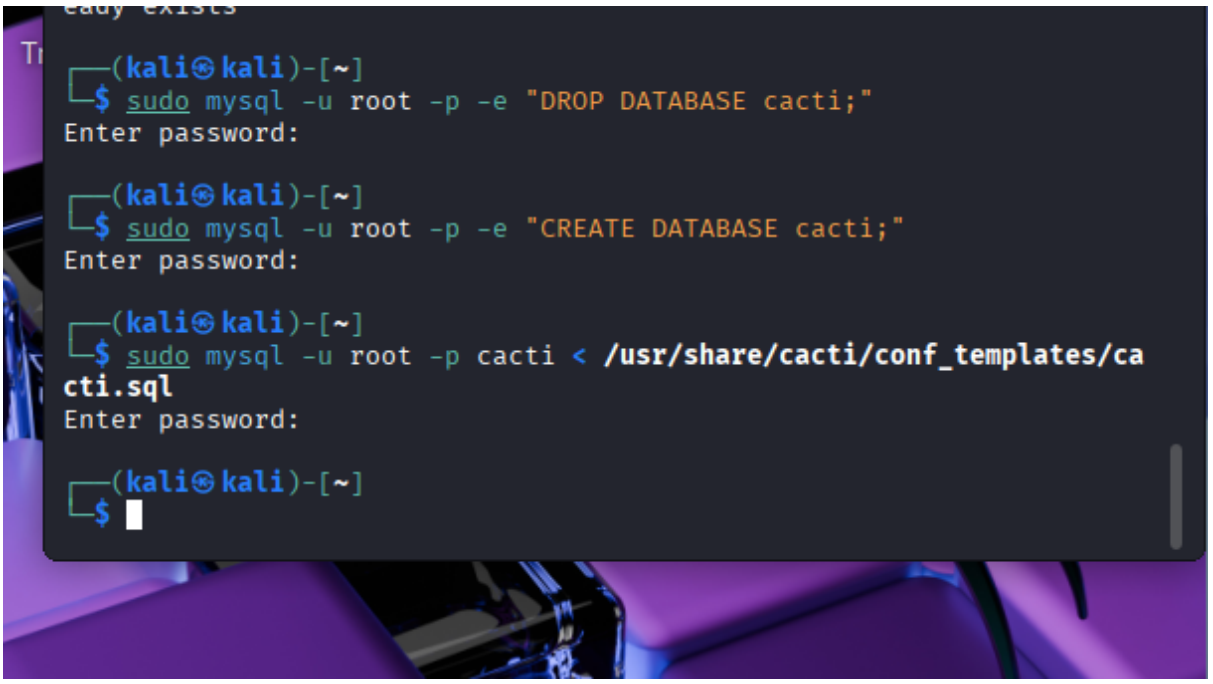
Importa la base cacti.sql dentro de la BD recién creada "cacti".

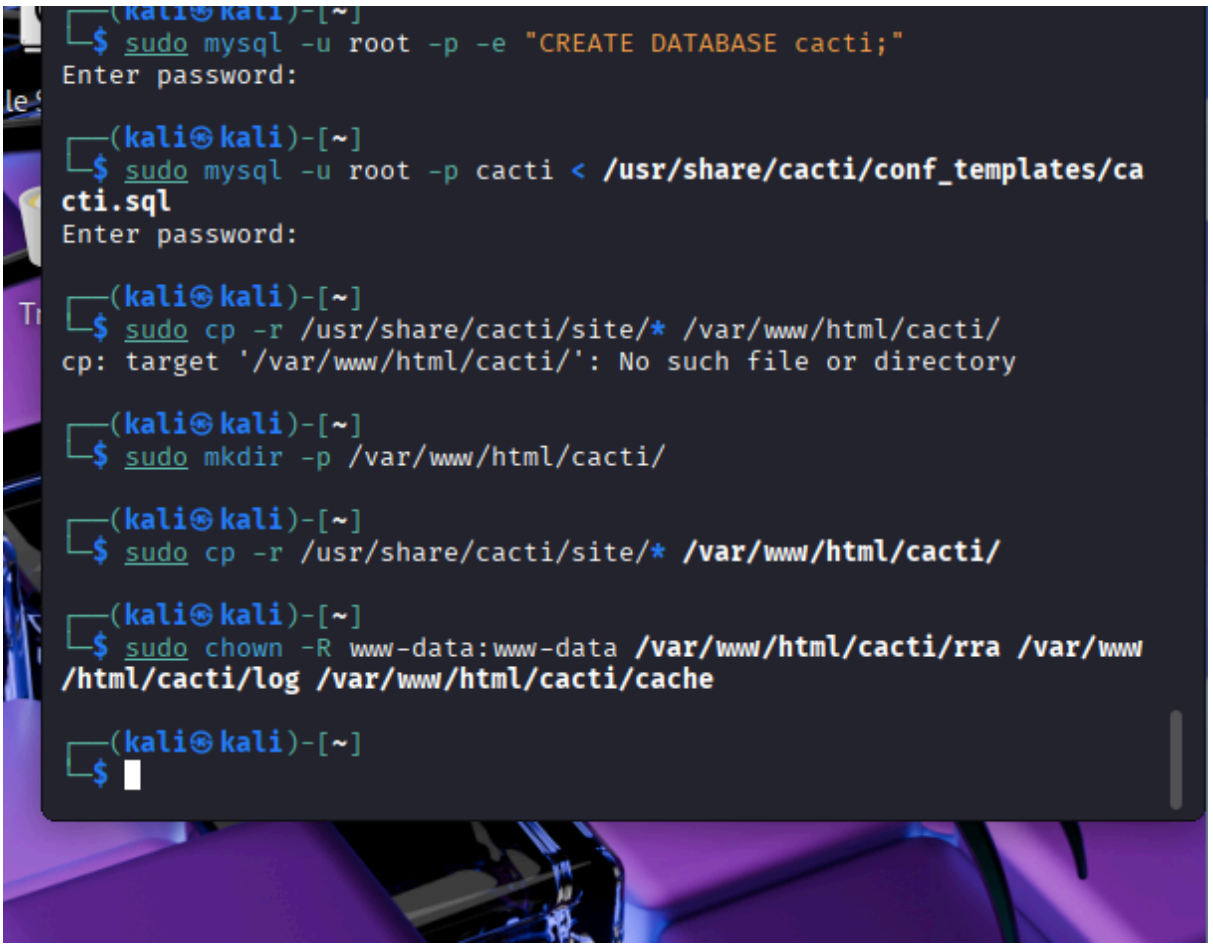
Solución

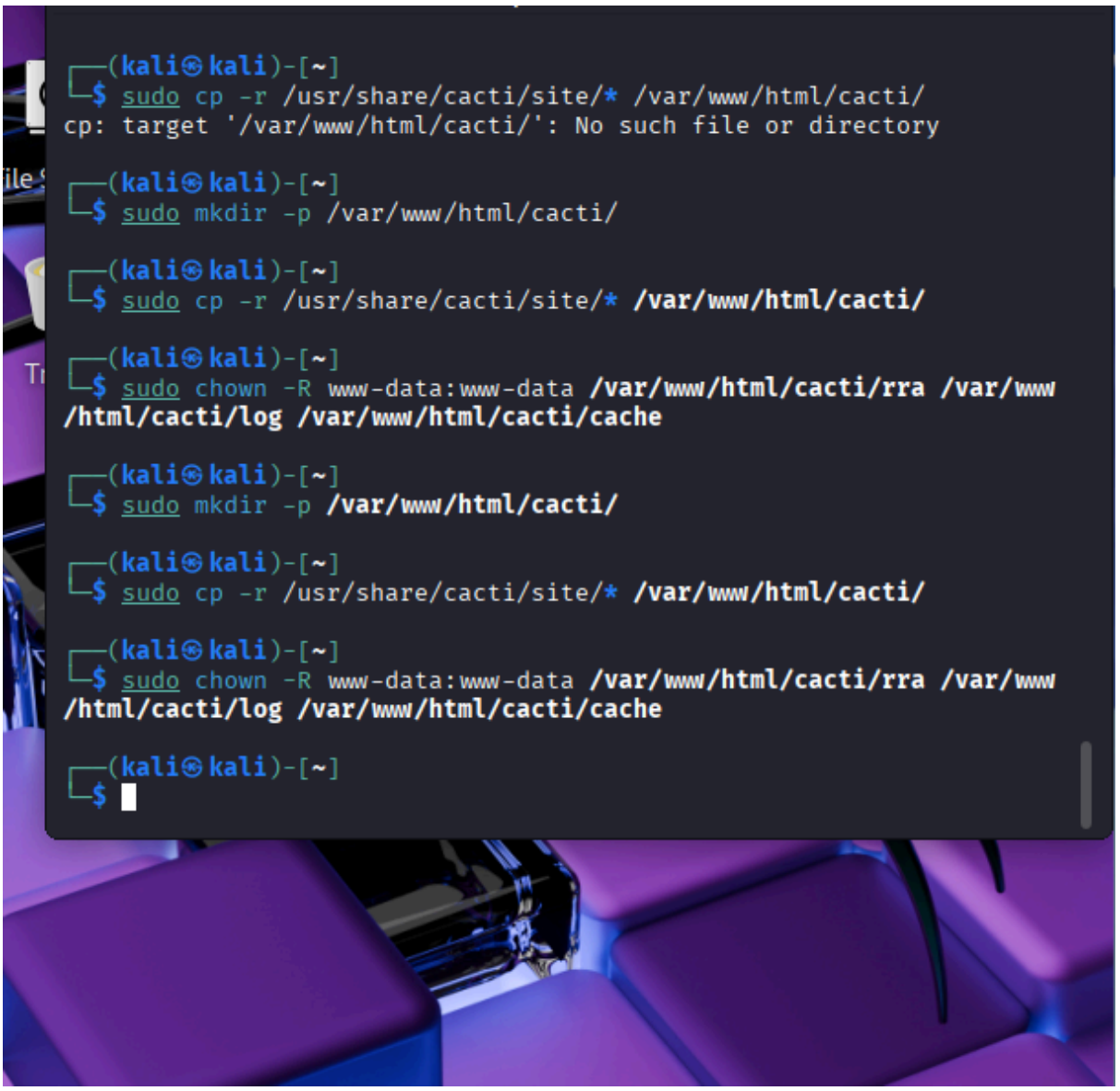
sudo mysql -u root -p cacti < /var/www/html/cacti/cacti.sql

Explicación

Es un paso obligatorio tras mover los archivos de Cacti al directorio web. Carga la estructura básica.

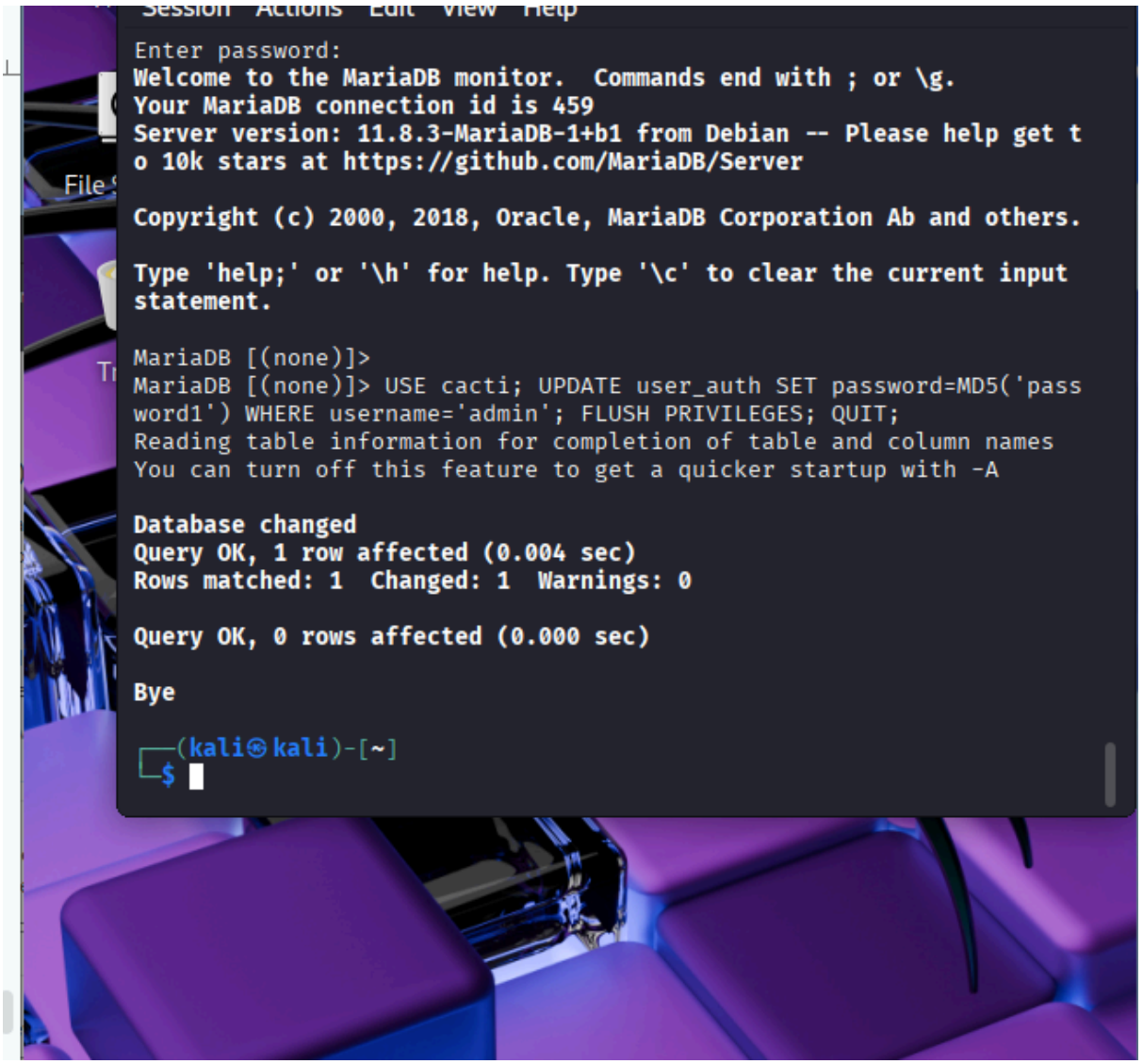

(kali㉿kali)-[~]
$ sudo mysql -u root -p -e "DROP DATABASE cacti;"
Enter password:
(kali㉿kali)-[~]
$ sudo mysql -u root -p -e "CREATE DATABASE cacti;"
Enter password:
(kali㉿kali)-[~]
$ sudo mysql -u root -p cacti < /usr/share/cacti/conf_templates/cacti.sql
Enter password:
(kali㉿kali)-[~]
$

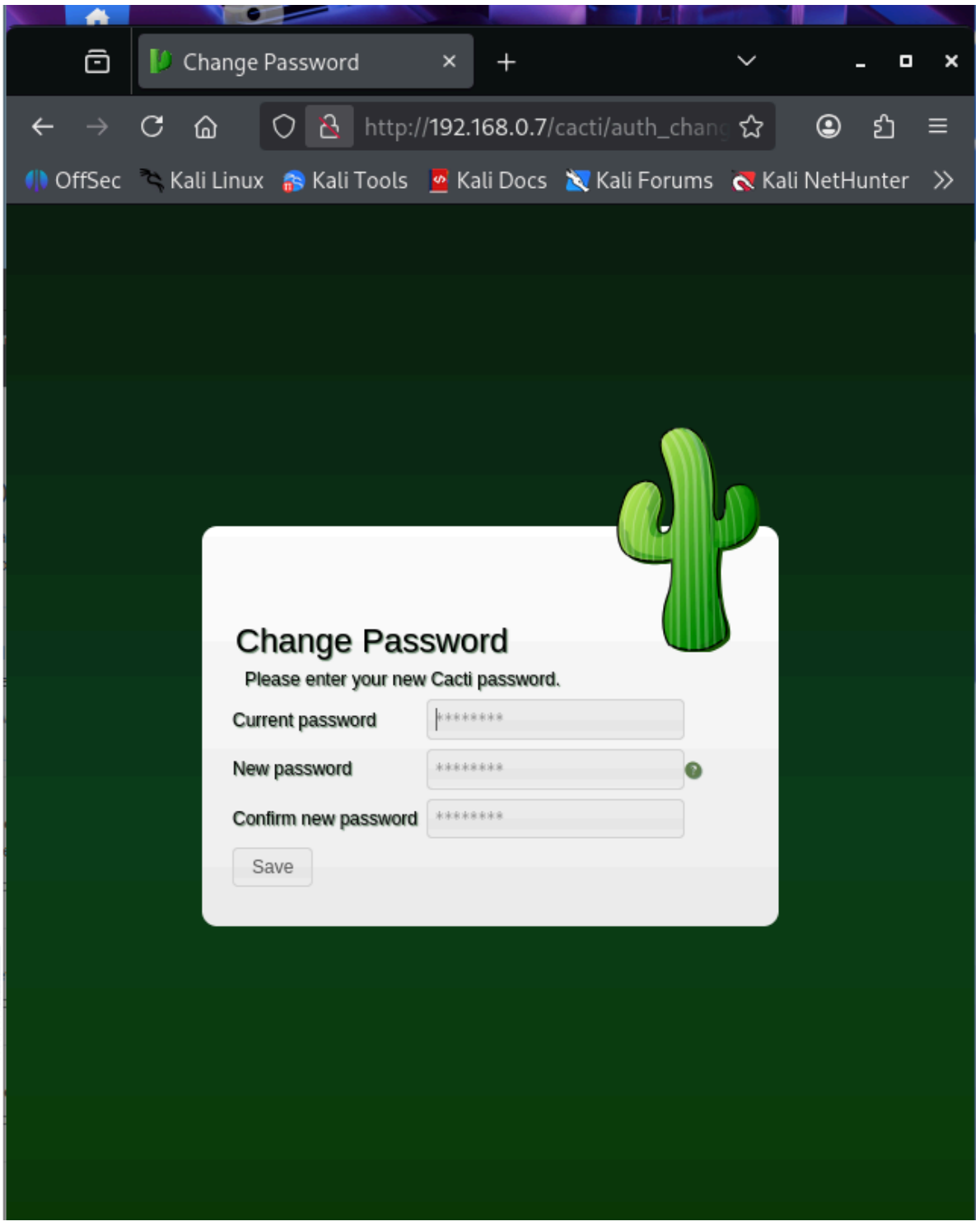

$ sudo mysql -u root -p -e "CREATE DATABASE cacti;"
Enter password:
(kali㉿kali)-[~]
$ sudo mysql -u root -p cacti < /usr/share/cacti/conf_templates/cacti.sql
Enter password:
(kali㉿kali)-[~]
$ sudo cp -r /usr/share/cacti/site/* /var/www/html/cacti/
cp: target '/var/www/html/cacti/': No such file or directory
(kali㉿kali)-[~]
$ sudo mkdir -p /var/www/html/cacti/
(kali㉿kali)-[~]
$ sudo cp -r /usr/share/cacti/site/* /var/www/html/cacti/
(kali㉿kali)-[~]
$ sudo chown -R www-data:www-data /var/www/html/cacti/rra /var/www/html/cacti/log /var/www/html/cacti/cache
(kali㉿kali)-[~]
$

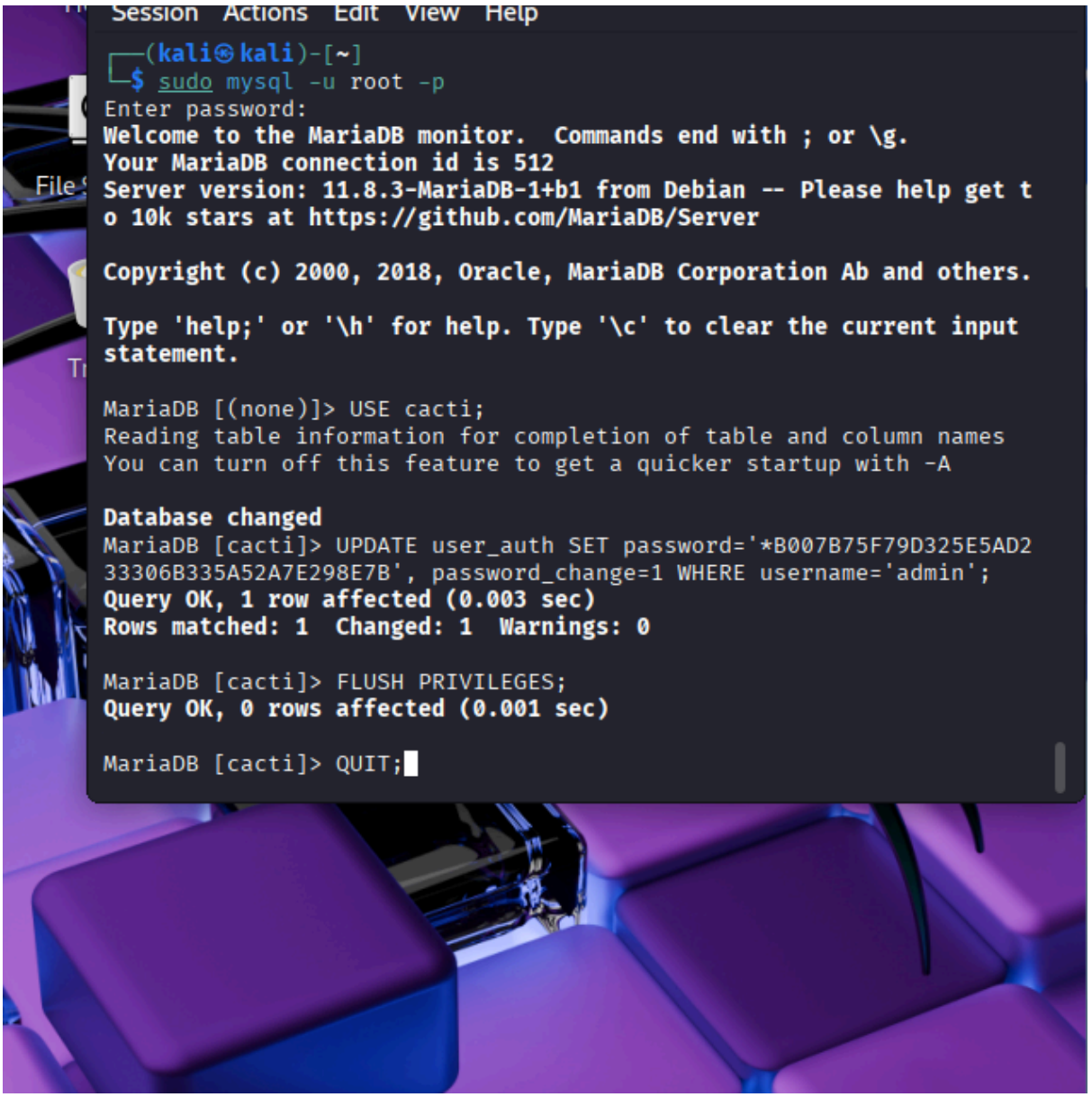


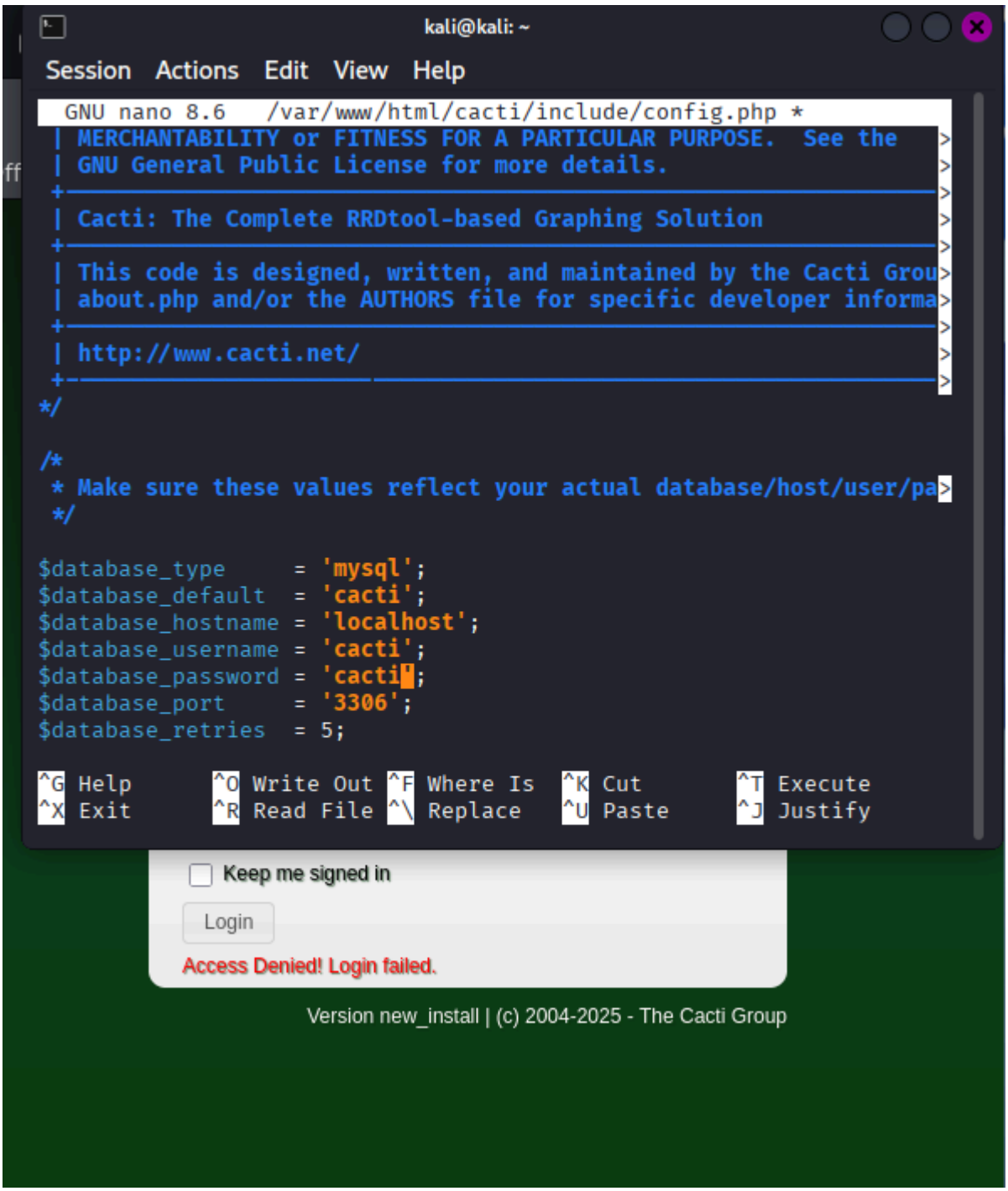


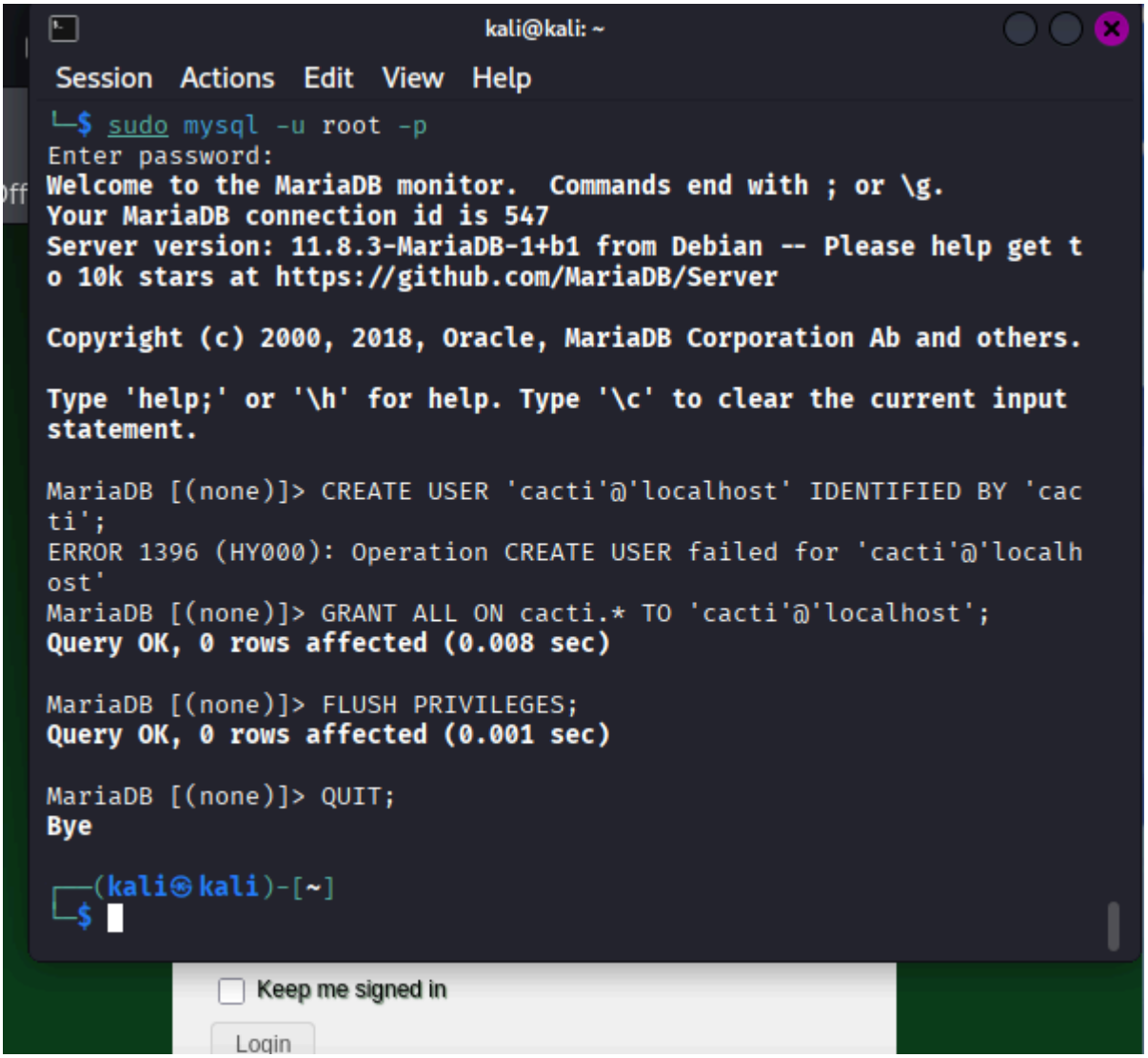
USE cacti; UPDATE user_auth SET password=MD5('password1') WHERE username='admin'; FLUSH PRIVILEGES; QUIT;


Session Actions Edit View Help
Enter password:
Welcome to the MariaDB monitor. Commands end with ; or \g.
Your MariaDB connection id is 459
Server version: 11.8.3-MariaDB-1+b1 from Debian -- Please help get t
o 10k stars at https://github.com/MariaDB/Server
Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.
Type 'help;' or '\h' for help. Type '\c' to clear the current input
statement.
MariaDB [(none)]>
MariaDB [(none)]> USE cacti; UPDATE user_auth SET password=MD5('pass
word1') WHERE username='admin'; FLUSH PRIVILEGES; QUIT;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A
Database changed
Query OK, 1 row affected (0.004 sec)
Rows matched: 1 Changed: 1 Warnings: 0
Query OK, 0 rows affected (0.000 sec)
Bye
(kali㉿kali)-[~]
$


Change Password
http://192.168.0.7/cacti/auth_chang
OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Change Password
Please enter your new Cacti password.
Current password
New password
Confirm new password
Save


Session Actions Edit View Help
(kali㉿kali)-[~]
$ sudo mysql -u root -p
Enter password:
Welcome to the MariaDB monitor. Commands end with ; or \g.
Your MariaDB connection id is 512
Server version: 11.8.3-MariaDB-1+b1 from Debian -- Please help get t
o 10k stars at https://github.com/MariaDB/Server
Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.
Type 'help;' or '\h' for help. Type '\c' to clear the current input
statement.
MariaDB [(none)]> USE cacti;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A
Database changed
MariaDB [cacti]> UPDATE user_auth SET password='*B007B75F79D325E5AD2
33306B335A52A7E298E7B', password_change=1 WHERE username='admin';
Query OK, 1 row affected (0.003 sec)
Rows matched: 1 Changed: 1 Warnings: 0
MariaDB [cacti]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.001 sec)
MariaDB [cacti]> QUIT;


kali@kali: ~
Session Actions Edit View Help
GNU nano 8.6 /var/www/html/cacti/include/config.php *
| MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE. See the
| GNU General Public License for more details.
| Cacti: The Complete RRDtool-based Graphing Solution
| This code is designed, written, and maintained by the Cacti Grou
| about.php and/or the AUTHORS file for specific developer informa
| http://www.cacti.net/
*/
/*
* Make sure these values reflect your actual database/host/user/pa
*/
$database_type = 'mysql';
$database_default = 'cacti';
$database_hostname = 'localhost';
$database_username = 'cacti';
$database_password = 'cacti';
$database_port = '3306';
$database_retries = 5;
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify
Keep me signed in
Login
Access Denied! Login failed.
Version new_install | (c) 2004-2025 - The Cacti Group



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4. Añadir host básico en Xymon (básico)

Enunciado

Añade un host llamado “kali-test” con IP 192.168.1.20 para testear ping.

Solución

Editar:

sudo nano /etc/xymon/hosts.cfg

Añadir:

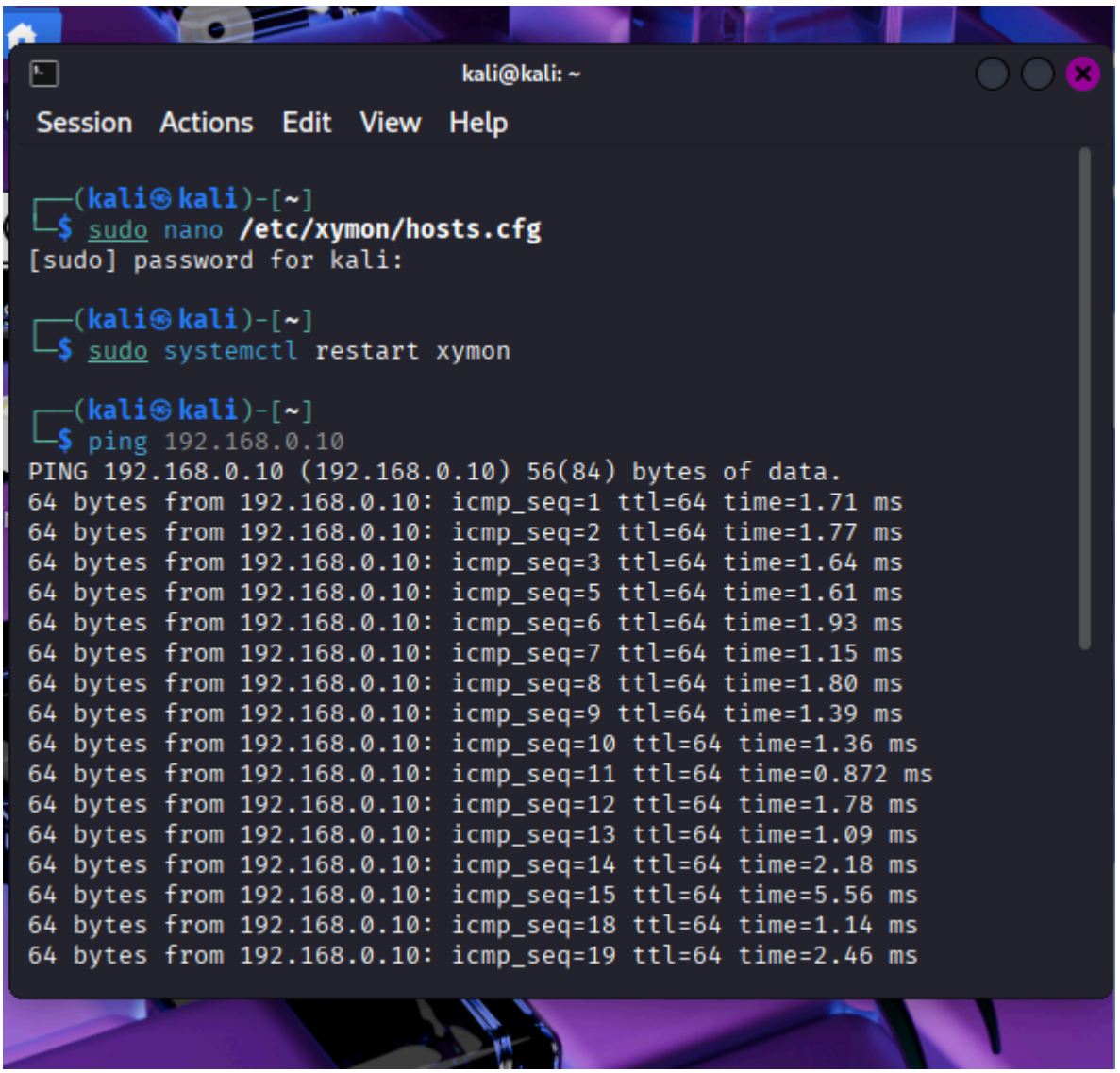
192.168.1.20 kali-test # testip

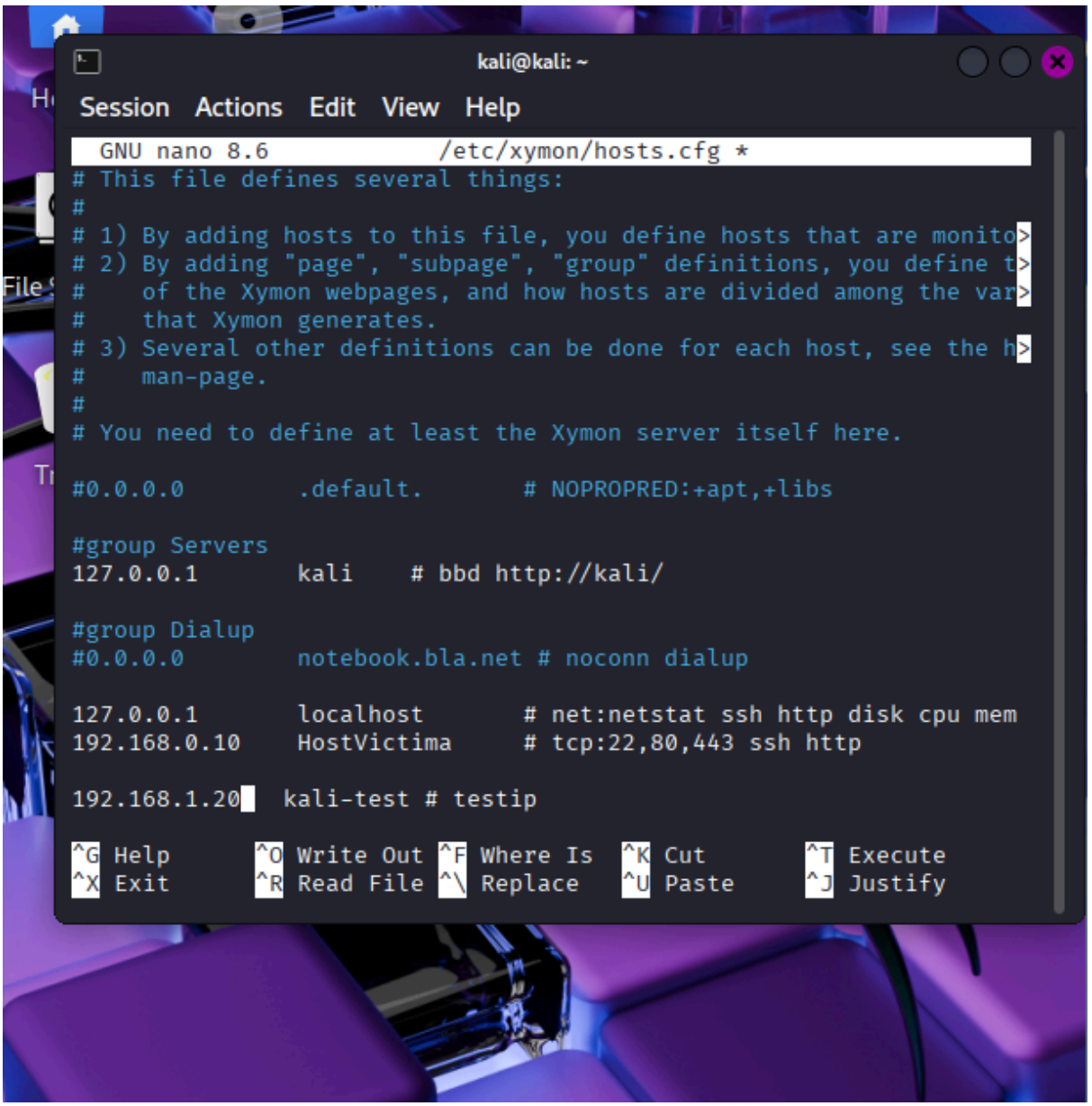
Reiniciar:

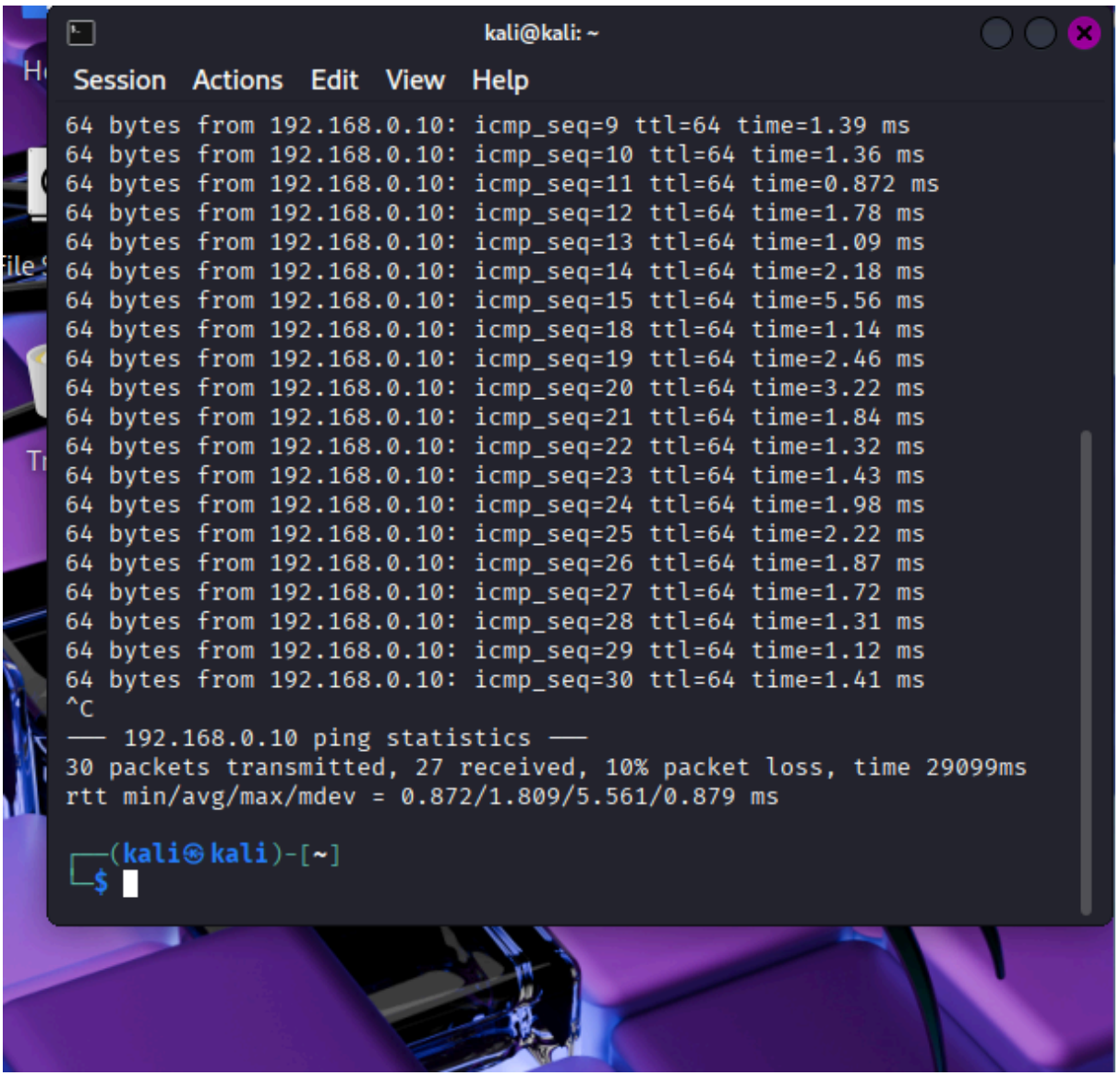
sudo systemctl restart xymon

Explicación

testip activa la comprobación de ping. Esto está documentado como ejemplo válido.




kali@kali: ~
Session Actions Edit View Help
GNU nano 8.6 /etc/xymon/hosts.cfg *
This file defines several things:
#
1) By adding hosts to this file, you define hosts that are monito>
2) By adding "page", "subpage", "group" definitions, you define t>
of the Xymon webpages, and how hosts are divided among the var>
that Xymon generates.
3) Several other definitions can be done for each host, see the h>
man-page.
#
You need to define at least the Xymon server itself here.
#0.0.0.0 .default. # NOPROPRED:+apt,+libs
#group Servers
127.0.0.1 kali # bbd http://kali/
#group Dialup
#0.0.0.0 notebook.bla.net # noconn dialup
127.0.0.1 localhost # net:netstat ssh http disk cpu mem
192.168.0.10 HostVictima # tcp:22,80,443 ssh http
192.168.1.20 kali-test # testip
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify



✅ EJERCICIOS DE NIVEL MEDIO (CACTI + XYMON)

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5. Configurar SNMP para Cacti de forma correcta (medio)

Enunciado

Modifica snmpd.conf para permitir que Cacti recoja métricas del propio sistema. ¿Qué líneas se deben cambiar?

Solución

sudo nano /etc/snmp/snmpd.conf

Cambiar/asegurar:

agentAddress udp:161,udp6:[::1]:161

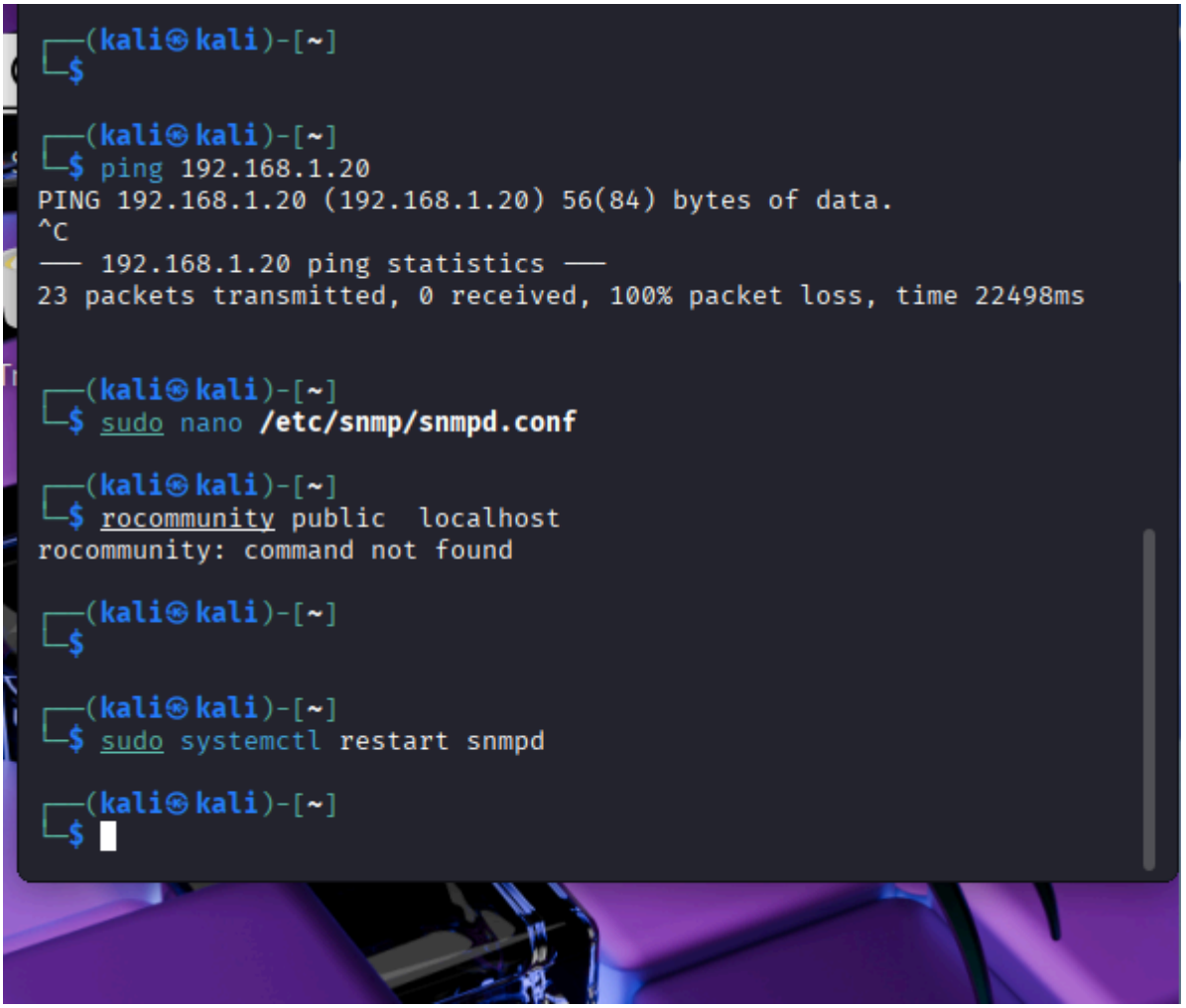
rocommunity public localhost

uego:

sudo systemctl restart snmpd

Explicación

Estas dos líneas permiten que SNMP responda localmente. El documento lo explica literalmente.



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6. Crear cron para poller de Cacti (medio)

Enunciado

Crea un cron que ejecute el poller de Cacti cada 2 minutos en vez de cada 5, como en el manual.

Solución

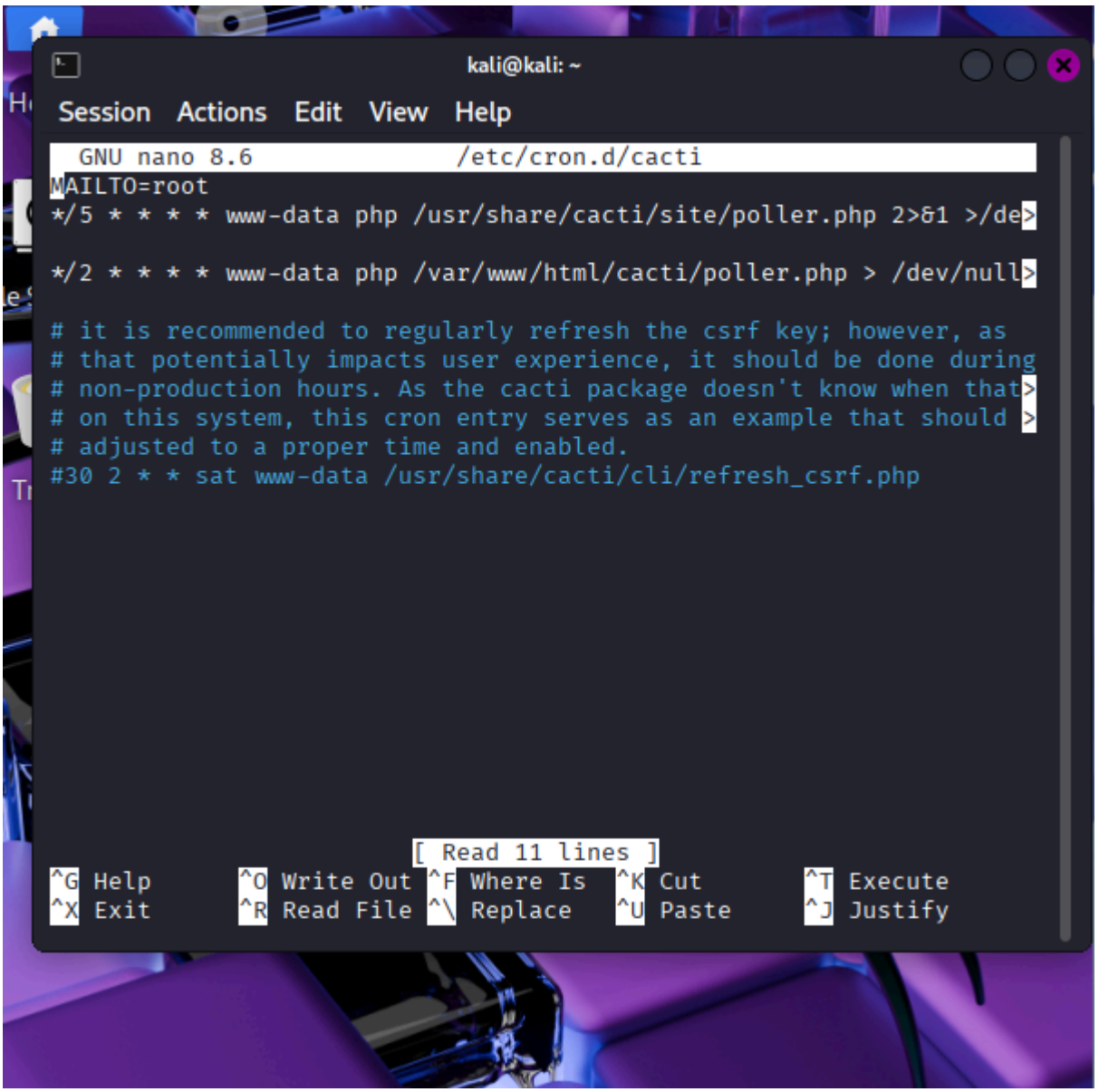
sudo nano /etc/cron.d/cacti

Añadir:

*/2 * * * * www-data php /var/www/html/cacti/poller.php > /dev/null 2>&1

Explicación

El cron permite que Cacti recoja datos periódicos; en tu manual la plantilla usa cada 5 minutos. Aquí se ajusta el intervalo.



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7. Comprobar el estado del servicio Xymon y logs (medio)

Enunciado

Comprueba que Xymon funciona correctamente y revisa sus logs en tiempo real.

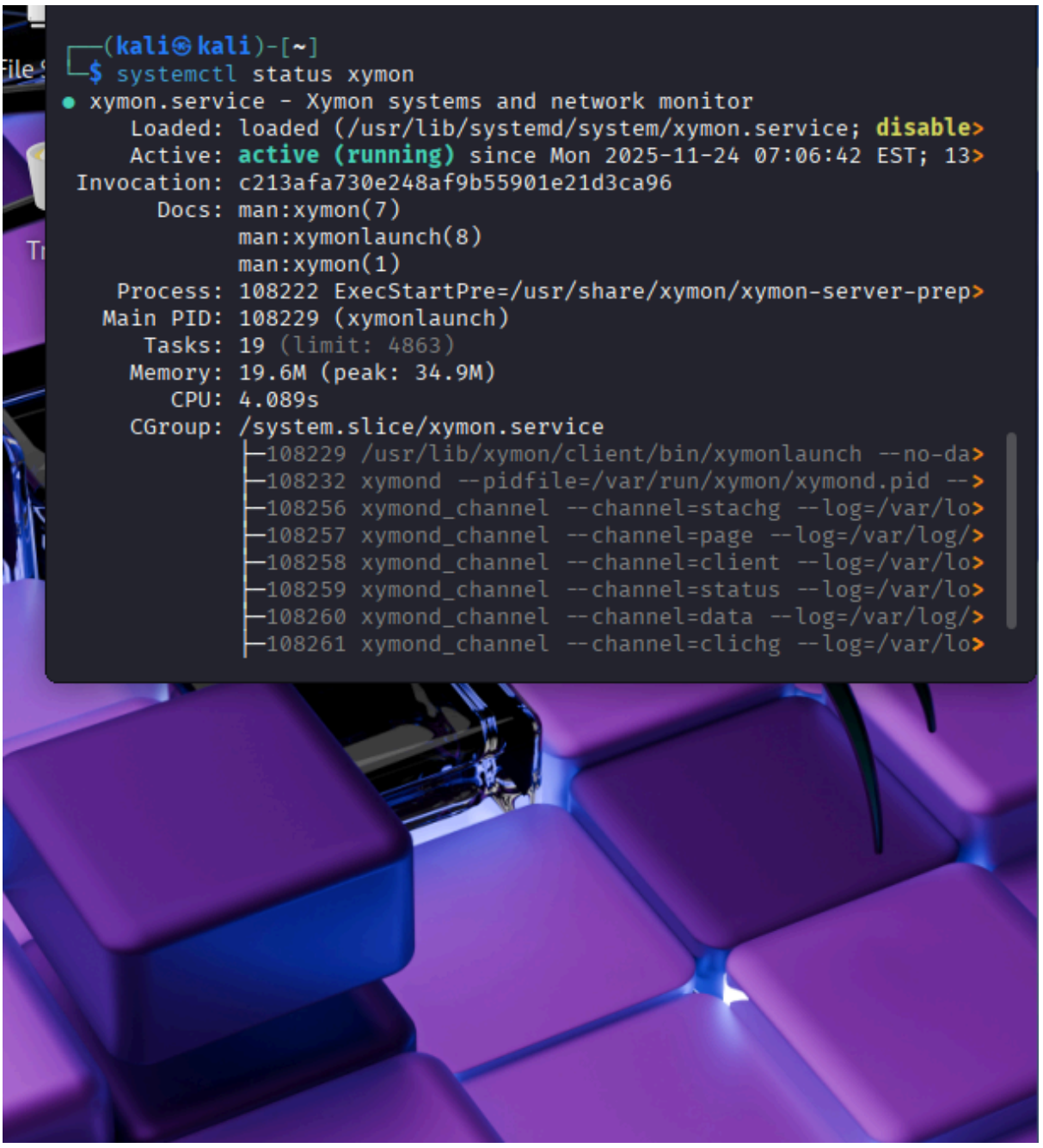
Solución

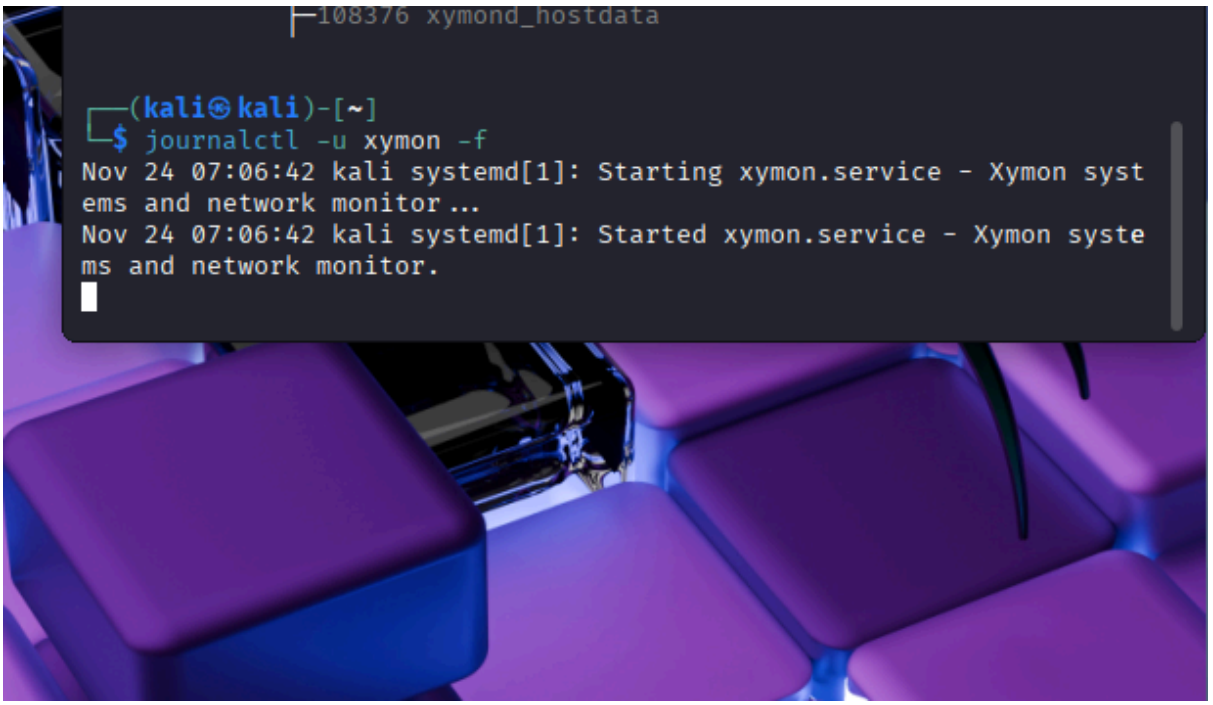
systemctl status xymon

journalctl -u xymon -f

Explicación

El documento recomienda usar journalctl -u xymon -f para ver errores del servicio.


(kali㉿kali)-[~]
$ systemctl status xymon
● xymon.service - Xymon systems and network monitor
Loaded: loaded (/usr/lib/systemd/system/xymon.service; disable>
Active: active (running) since Mon 2025-11-24 07:06:42 EST; 13>
Invocation: c213afa730e248af9b55901e21d3ca96
Docs: man:xymon(7)
man:xymonlaunch(8)
man:xymon(1)
Process: 108222 ExecStartPre=/usr/share/xymon/xymon-server-prep>
Main PID: 108229 (xymonlaunch)
Tasks: 19 (limit: 4863)
Memory: 19.6M (peak: 34.9M)
CPU: 4.089s
CGroup: /system.slice/xymon.service
├─108229 /usr/lib/xymon/client/bin/xymonlaunch --no-da>
├─108232 xymond --pidfile=/var/run/xymon/xymond.pid -->
├─108256 xymond_channel --channel=stachg --log=/var/lo>
├─108257 xymond_channel --channel=page --log=/var/log/>
├─108258 xymond_channel --channel=client --log=/var/lo>
├─108259 xymond_channel --channel=status --log=/var/lo>
├─108260 xymond_channel --channel=data --log=/var/log/>
├─108261 xymond_channel --channel=clichg --log=/var/lo>



---

8. Añadir un host completo en Xymon con tests SSH y HTTP (medio)

Enunciado

Añade un equipo “servidor-web” con IP 10.0.2.25 y habilita los tests:

- Ping (testip)
- Conectividad TCP (conn)
- SSH
- HTTP

Solución

Editar hosts.cfg:

sudo nano /etc/xymon/hosts.cfg

Añadir:

10.0.2.25 servidor-web # testip conn ssh http

Reiniciar:

sudo systemctl restart xymon

Explicación

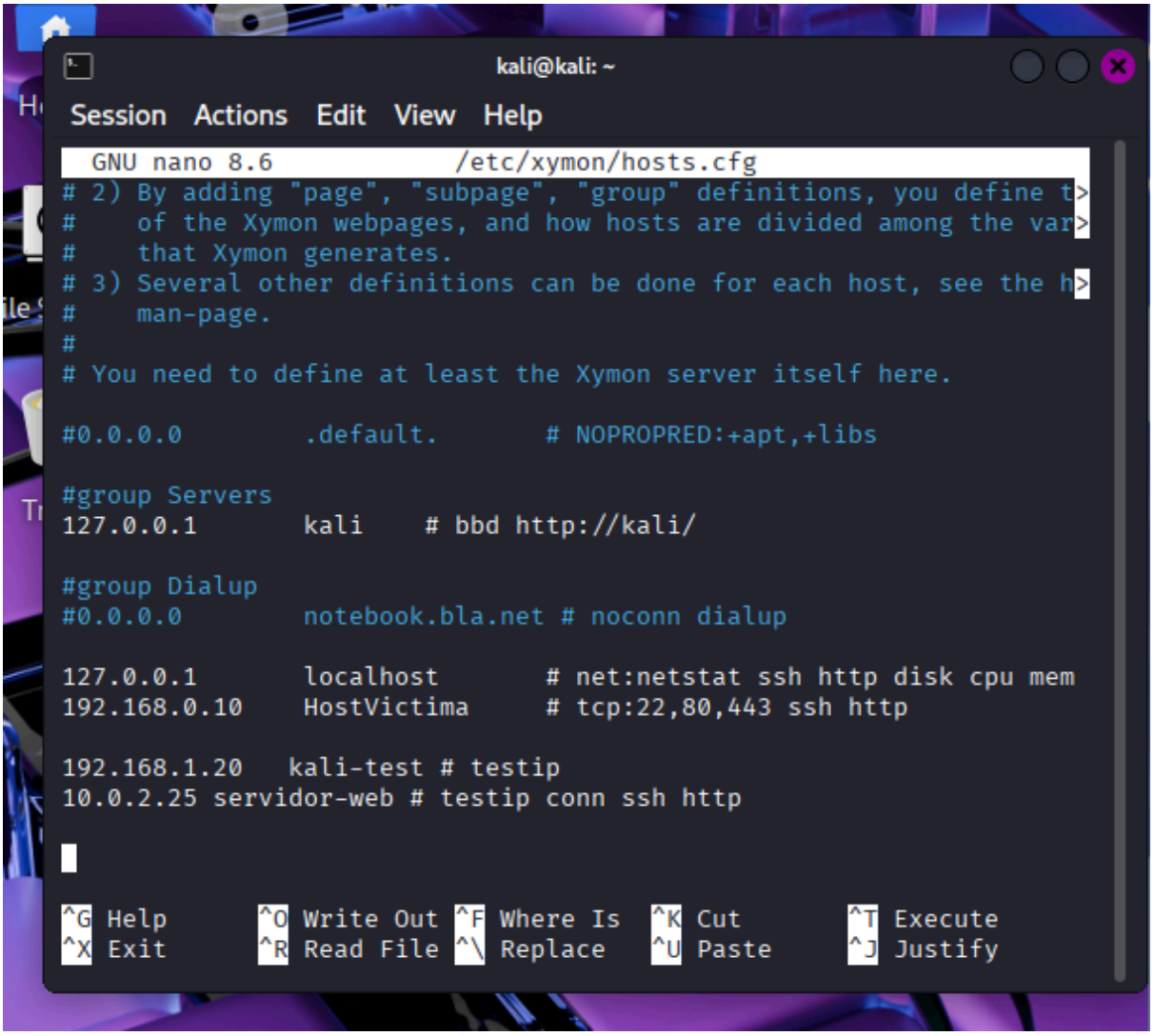
Los tests están definidos tal como describe el documento, y se pueden activar junto al host.

```
┌──(kali㉿kali)-[~]
└─$ sudo nano /etc/xymon/hosts.cfg

┌──(kali㉿kali)-[~]
└─$ sudo systemctl restart xymon

┌──(kali㉿kali)-[~]
└─$ sudo nano /etc/xymon/hosts.cfg

┌──(kali㉿kali)-[~]
└─$
```



9. Configurar completamente config.php de Cacti (medio)

Enunciado

Edita el archivo config.php de Cacti y escribe los valores correctos para la BD llamada cacti, con el usuario cactiuser y contraseña 1234.

Solución

sudo nano /var/www/html/cacti/include/config.php

Modificar:

$database_type = "mysql";

$database_default = "cacti";

$database_hostname = "localhost";

$database_username = "cactiuser";

$database_password = "1234";

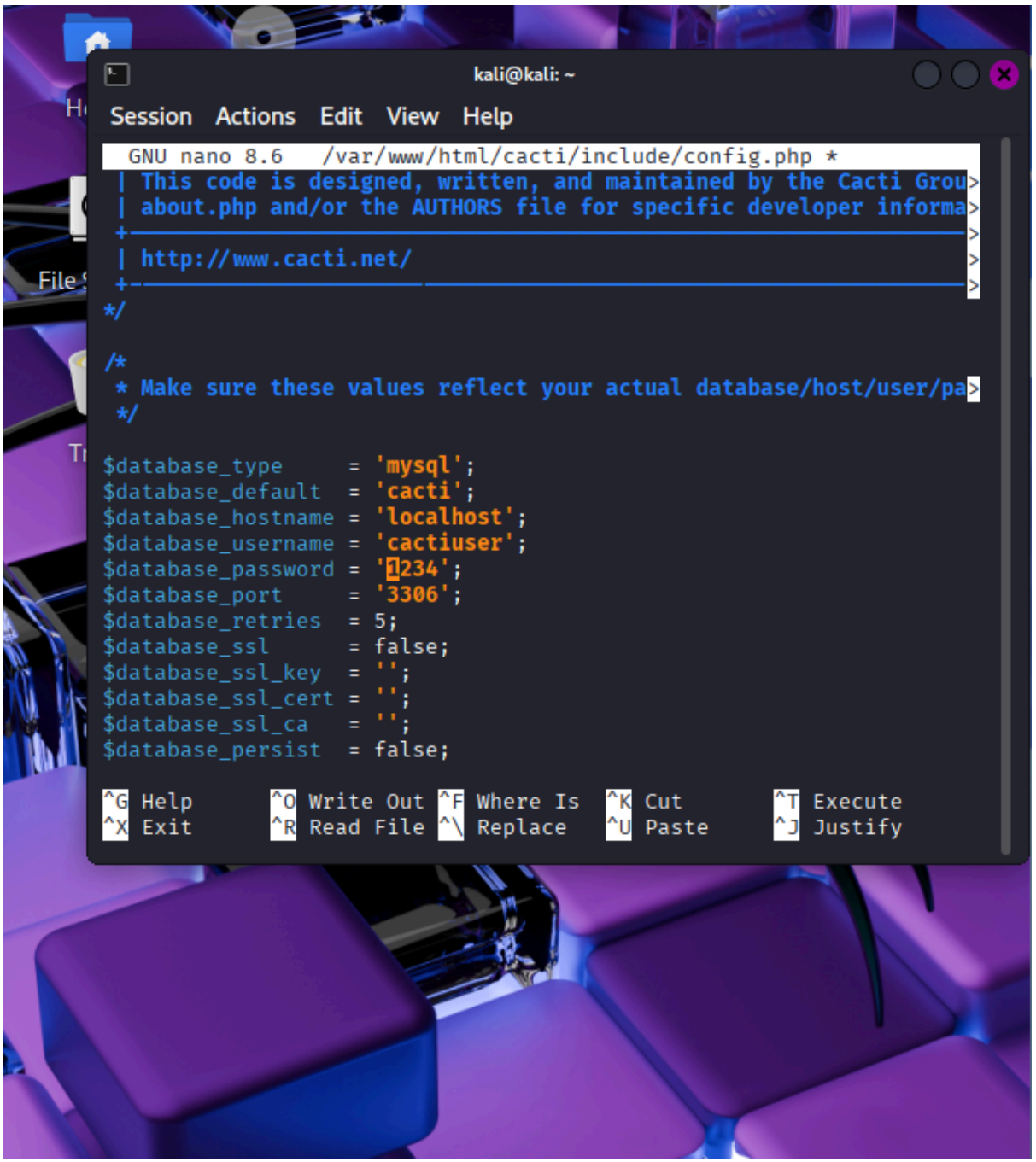
$database_port = "3306";

sudo nano /usr/share/cacti/site/include/config.php

sudo systemctl restart apache2

Explicación

Estos parámetros son imprescindibles para que Cacti acceda a MariaDB. Aparecen en tu documento de instalación.



---

10. Habilitar módulos de Apache necesarios para Xymon (medio)

Enunciado

Xymon necesita algunos módulos de Apache. ¿Cuáles son y cómo se habilitan?

Solución

sudo a2enmod cgi

sudo a2enmod cgid

sudo a2enmod auth_basic

sudo a2enmod auth_digest

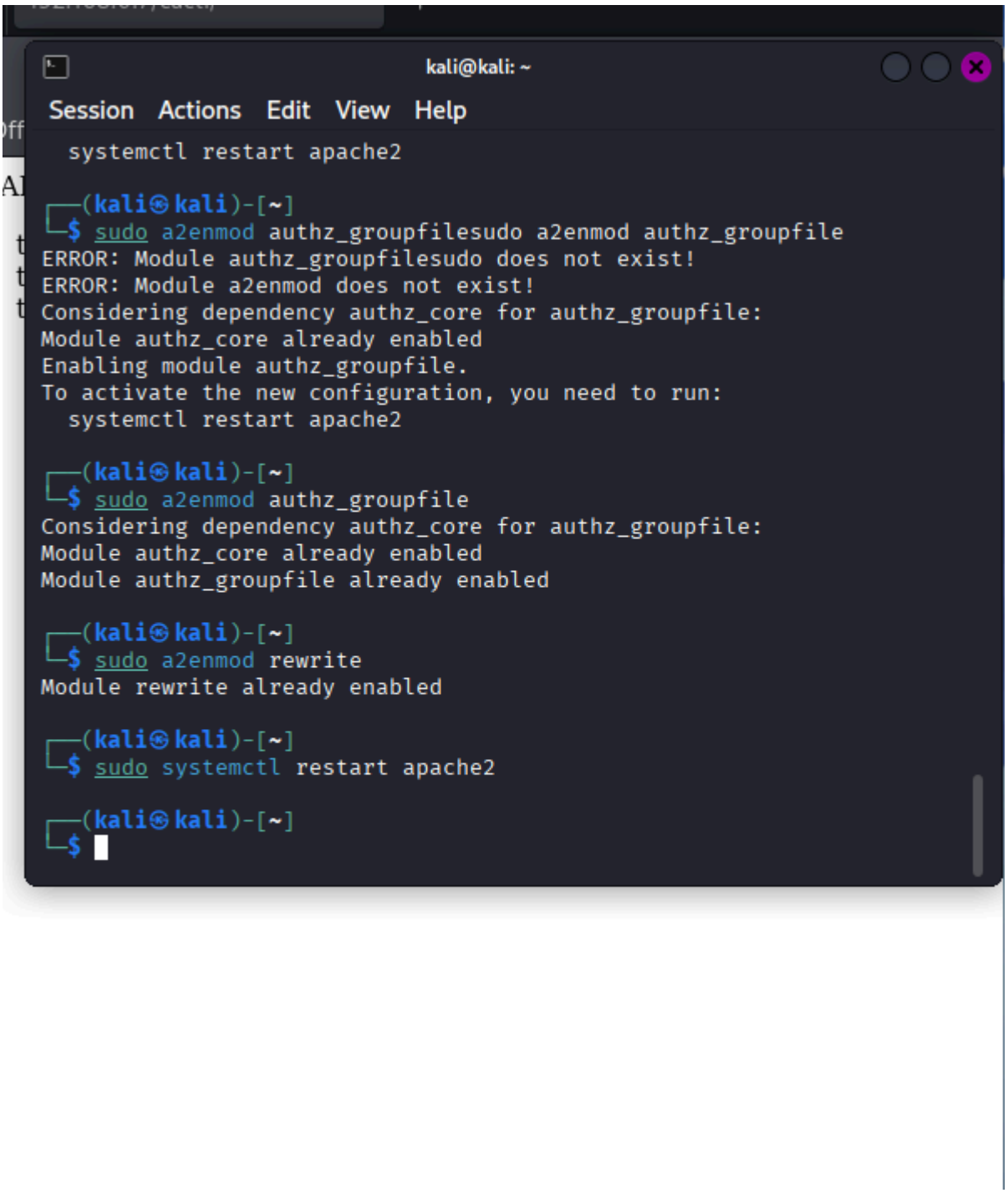
sudo a2enmod authz_groupfile

}sudo a2enmod rewrite

sudo systemctl restart apache2

Explicación

Tu script y manual indican exactamente estos módulos como necesarios para procesar las páginas CGI de Xymon.


kali@kali: ~
Session Actions Edit View Help
systemctl restart apache2
(kali㉿kali)-[~]
$ sudo a2enmod authz_groupfilesudo a2enmod authz_groupfile
ERROR: Module authz_groupfilesudo does not exist!
ERROR: Module a2enmod does not exist!
Considering dependency authz_core for authz_groupfile:
Module authz_core already enabled
Enabling module authz_groupfile.
To activate the new configuration, you need to run:
systemctl restart apache2
(kali㉿kali)-[~]
$ sudo a2enmod authz_groupfile
Considering dependency authz_core for authz_groupfile:
Module authz_core already enabled
Module authz_groupfile already enabled
(kali㉿kali)-[~]
$ sudo a2enmod rewrite
Module rewrite already enabled
(kali㉿kali)-[~]
$ sudo systemctl restart apache2
(kali㉿kali)-[~]
$